Nifty overnight timing and NSE momentum

Prathama Bharadwaj. Quant Research Intern take-home, H TO H Partners LLP. All numbers are from *harness_public.py*, after costs, unless stated.

	Sharpe	Max DD	CAGR	Trades/yr	Bar to beat	Firm reference: Sharpe, DD, CAGR	My expectation on hidden data
Task 1, Apr 2017 to Jan 2022	2.95	-4.7%	22.5%	110	Sharpe > 1.0, DD better than -15%	1.22, -12.8%, 23.3%	Sharpe about 1, DD within -10%
Task 2, 2 Mar 2022 to 30 Jun 2024	3.05	-11.2%	48.0%	94	Sharpe > 1.9, DD better than -25%	1.93, -23.9%, 51.6%	Sharpe 1 to 1.5, DD within -15%

Both clear their bars by a wide margin, but both in-sample windows flatter them: Task 1 through an overnight premium that shrank after 2021, Task 2 through a universe where 15 random stocks already score Sharpe 1.66. The stress and out-of-sample results below are the better guide, and they are where my expectations come from.

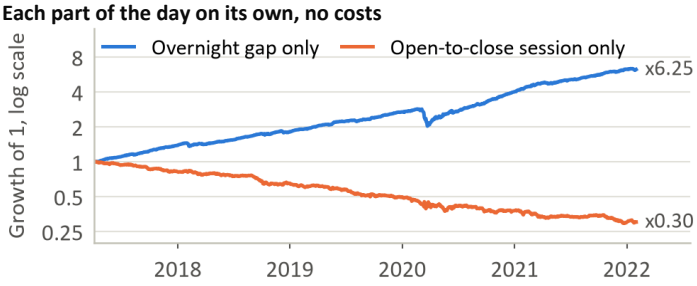
Task 1: long overnight, in healthy markets only

Idea. Nifty's return arrives while the market is shut. From 2017 to Jan 2022 the overnight gap has a Sharpe of +3.4 (t = 7.5) and the open-to-close session -1.6 (t = -3.4), with those signs in every full year. The Task 2 daily file confirms it past the 15-minute data: +2.6 and -0.8 for Feb 2022 to Jun 2024. The 09:15 bar alone loses 6.5 bps a day (t = -6.2).

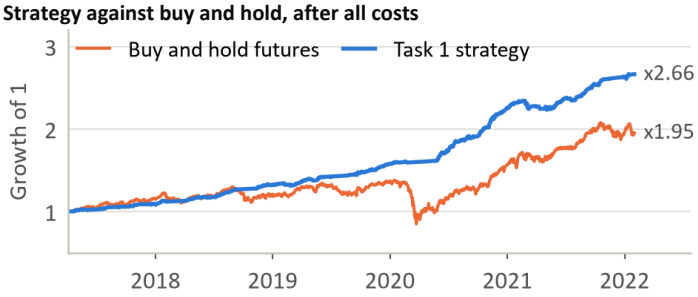
Why it should work. Overnight holders carry gap risk that is hard to hedge or exit while the cash market is shut, and intraday leverage is normally closed by the end of the day, so overnight capital should be scarce and get paid. Buying pressure at the open then lifts the first print, which fades (Berkman et al. 2012; Lou, Polk and Skouras 2019). The premium is thin and the tail is large when the market falls, so I only take it when the market is calm and rising.

Rules. At the close of the 14:30 bar, go long if (1) price is above its 20-day average, (2) the session so far is at least -0.25%, (3) 20-day realised vol is under 35%, and (4) the last 120 paper trades averaged at least 7 bps, which is the round-trip cost plus carry. Fill at 14:45, exit at the next morning's open, otherwise stay flat. Never short: shorts pay the gap, and long-short lost to long-flat in 10 of 11 trend configs. Deciding at 14:30 still fills before the close with a 2-bar delay. Rule 4 never fired in-sample, so it cannot have been fitted to this data. It is there in case the premium fades.

Each part of the day on its own, no costs



Strategy against buy and hold, after all costs



Results and stress cases

Case	CAGR	Max DD	Sharpe
Official, Apr 2017 to Jan 2022	**22.5%**	**-4.7%**	**2.95**
1-bar fill delay	21.4%	-4.7%	2.61
2-bar fill delay	16.9%	-4.4%	1.94
2x costs	15.6%	-5.4%	1.78
Best year (2020) removed	18.2%	-4.7%	2.37
2017 to 2019	18.1%	-2.5%	2.70
2020 to Jan 2022	28.5%	-4.7%	3.27
Out-of-sample, Feb 22 to Jun 24	14.8%	-5.4%	1.46 (0.51 to 2.53)

Year	2017 (Apr on)	2018	2019	2020	2021	2022 (Jan)
Return	9.2%	21.2%	19.2%	42.6%	16.6%	1.5%

Turnover and cost drag. 110 round trips a year, long on 45% of nights and 5% of bars. Zero-cost CAGR is 29.8%, so trading and roll costs take 7.3 points a year. Win rate 69%, mean trade 15.3 bps, worst trade -1.79%. Top 10 trades are 19% of net profit.

Out-of-sample check. The 15-minute data stops in Jan 2022, so I built later bars from the real daily OHLC with seeded synthetic paths inside each day. Gaps and ranges are real, so this only tests day-level logic. On real in-sample days it landed within about 0.6 of the true Sharpe, usually below. It is not a clean holdout, because I chose the session floor after seeing it. Across the 192 filter combinations the median is 1.12, which is why I expect about 1. Bootstrap 95% CI on the official Sharpe: 1.81 to 3.98. Deflated Sharpe ratio for 381 logged configs: 0.93.

Sensitivity: in-sample Sharpe | out-of-sample Sharpe

MA days \ session floor	-0.75%	-0.50%	-0.25%	0.00%
10	2.87 | 0.67	2.88 | 0.79	2.74 | 1.13	2.71 | 1.27
15	3.01 | 0.85	3.03 | 0.96	2.87 | 1.17	2.72 | 1.12
20	3.12 | 0.98	3.08 | 0.97	2.95 | 1.43	2.89 | 1.42
25	3.03 | 1.00	3.01 | 1.02	2.88 | 1.47	2.78 | 1.39
30	3.04 | 0.82	3.05 | 0.79	2.91 | 1.26	2.79 | 1.31

All 20 cells beat the bar, with max DD between -7.9% and -2.9%.

Parameters and ranges tested

Parameter	Value	Range tested	Sharpe over range
ma_days	20	5 to 200	2.28 to 2.72, no other filters
decide_at	14:30	13:30 to 15:00	2.72 to 2.95 up to 14:30; later fails the 2-bar delay
session_floor	-0.25%	-1% to 0	2.89 to 3.12
vol_cap	35%	15% to off	2.71 to 2.95 from 25% up
gate_nights	120	60 to 250	2.90 to 2.95
gate_bps	7	off to 15	2.71 to 2.95
Vol window*	20 days	10, 20	2.88, 2.89, no floor or gate

*Module constant, beyond the six in PARAMS.

What would break it

The premium fading. Removing 0, 3, 6, 9 or 12 bps from every overnight gap gives out-of-sample Sharpe 1.43, 0.95, -0.11, -0.29, -0.68, and in-sample 2.95, 2.14, 1.63, 1.12, 0.65. The real out-of-sample gap averages 9.4 bps, so losing 6 bps makes it unprofitable. **Fills.** The harness credits the spot open in full, but futures may not reach that print. The 1-bar delay, which exits at 09:30, is my best proxy, and it holds at 2.61. **Costs.** Sharpe is 0.60 at 3x and -0.57 at 4x, so all-in round trips above about 17 bps end it. **A large gap with the filters on.** The worst trade was -1.79%, but filters cannot see overnight news, and Nifty opened 9% down in Mar 2020.

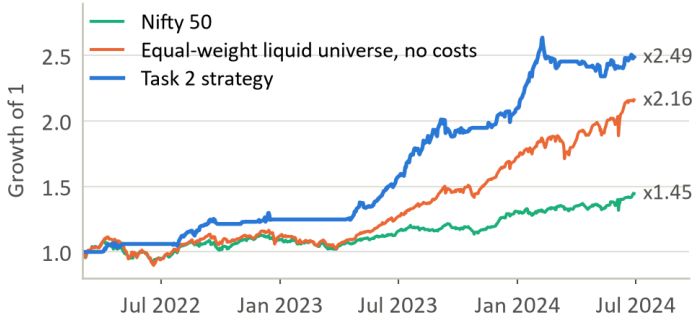
Task 2: momentum, in broad markets only

Idea. Own the 15 stocks with the best return per unit of risk, but only while most of the market is above trend. News diffuses slowly through mid and small caps, which is why momentum pays, but its edge depends on the market: the top 15 by 12-month momentum lagged the universe by 3.4% a quarter in 2022 and beat it by 13% in 2023. The filter keeps the strategy out of the first kind of market.

Rules. Each close: (1) if under 60% of liquid stocks (Rs 10 cr median traded value) are above their 50-day average, sell everything and buy nothing. (2) Rank liquid stocks by mean over std of 126-day returns, excluding vol above 65%. (3) Every 10 days, sell holdings ranked below 40. (4) Fill free slots with stop orders on the best-ranked names at their 10-day high, 1/15 of equity each. No stop-losses: they cut Sharpe by up to 0.25 and did not reduce drawdown.

Corporate actions. One-day moves beyond 45% (SPLPETRO -49%, and NMDC +86%, which the README does not list) are zeroed in all indicators, held names are sold at the next open, and new buys are blocked for 10 days. A -50% shock injected into a held name cost 0.22 of Sharpe.

Growth of 1 from 2 Mar 2022



Results and stress cases

Case	CAGR	Max DD	Sharpe
Official, 2 Mar 2022 to 30 Jun 2024	**48.0%**	**-11.2%**	**3.05**
Mean of 4 rebalance offsets	46.6%	-10.9%	2.97
2x costs	44.2%	-12.7%	2.85
Best year (2023) removed	35.4%	-11.2%	2.43
Random 70% of stocks, 20 draws	43.2%	-10.7%	2.84 (min 2.31)
Liquidity floor Rs 50 cr, not Rs 10 cr	27.6%	-13.4%	1.92
Tough: universe about +10% a year, not +47%	16.1%	-7.8%	1.57
Bear: universe about -10% a year	0.9%	-7.0%	0.20
Oct 2021 to Feb 2022, before the window*	0.7%	-2.8%	0.44

*Total return, not CAGR, and not a scored result, as the brief reserves 2021 for indicator warm-up. It did inform my design choices: see Checks on page 3. The liquidity floor, tough and bear rows are means over 4 offsets.

Year	2022 (Mar on)	2023	2024 (to Jun)
Return	24.8%	67.3%	19.2%

Turnover and cost drag. 94 trades a year in 130 names, 14.7x turnover, mean hold 19 days, in the market 50% of the time. Zero-cost CAGR is 51.9%, so costs take 3.9 points a year.

Top 10 trades: 57% of net profit (40% of gross profit), and the top 5 names are 40% of net. This concentration in a few multi-baggers is the weakest feature of the result. Random 70% sub-universes still average 2.84, so it does not rest on any one name.

What each layer adds

Same rules	Sharpe	CAGR	Max DD	Tough Sharpe
Random picks, no filter	1.73 (std 0.36)	34.4%	-19.6%	0.09
Random picks + breadth filter	1.78 (std 0.34)	19.8%	-11.1%	0.54
Selection, no filter	2.20	59.8%	-21.3%	0.92
Selection + filter, submitted	**2.97**	**46.6%**	**-10.9%**	**1.57**

Random rows use one fixed random ranking per seed, 24 seeds.

The filter alone mostly halves drawdown. Selection alone adds return. Together they are worth more, which fits momentum paying best in healthy markets. Bootstrap 95% CI on the official Sharpe: 1.56 to 4.64. Deflated Sharpe ratio for 1,396 logged configs: 0.99, which says nothing about universe bias.

Sensitivity: Sharpe | tough-case Sharpe, mean of 4 offsets

Breadth min \ lookback days	84	126	189
0.50	2.39 | 1.18	2.58 | 1.75	2.44 | 1.15
0.55	2.04 | 1.01	2.27 | 1.62	2.30 | 1.27
0.60	2.46 | 0.97	2.97 | 1.57	2.68 | 1.35
0.65	2.48 | 0.80	2.79 | 1.48	2.56 | 1.38
0.70	2.68 | -0.40	3.33 | 0.78	2.99 | 0.53

All cells beat the bar, with max DD between -15.5% and -6.6%. Neighbouring cells differ by up to about 0.7, so treat smaller gaps as noise. The 0.70 row has the best in-sample Sharpe, but I rejected it because it would rarely invest in a lower-drift market, as the tough case shows.

Parameters and ranges tested, mean of 4 offsets

Parameter	Value	Range tested	Sharpe over range
lookback	126 days	84 to 252 days	2.46 to 2.97, 84 to 189 days
breadth_min	0.60	0 to 0.80	1.83 to 3.33
top_k	15	8 to 15	2.89 to 3.01
sell_rank	40	15 to 90	2.96 to 3.06
max_vol	65%	35% to off	2.50 to 2.97, from 45% up
breakout_days	10	1 to 40	2.91 to 3.02
Review interval*	10 days	5 to 21 days	2.87 to 3.05
Liquidity floor*	Rs 10 cr	Rs 10 to 200 cr	1.64 to 2.97
Breadth MA*	50 days	20 to 200 days	2.41 to 2.97, up to 100 days
Vol window*	120 days	60 to 250 days	2.76 to 2.97
Jump limit*	45%	30% to off	2.97 to 2.98
Jump cool-off*	10 days	0 to 40 days	2.97, no effect

*Module constant, beyond the six in PARAMS.

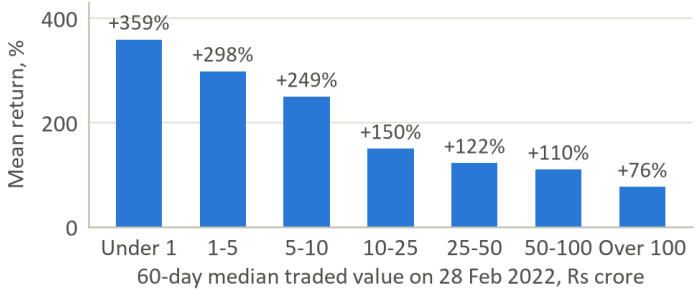
What would break it

Survivorship is the largest risk, covered on the next page. **Slow reaction.** The filter needs speed: if the signal arrives 1, 2, 3 or 10 days late, Sharpe is 2.85, 2.50, 1.77 and 1.60. **Choppy markets.** Breadth hovering near 60% means selling lows and rebuying highs at 0.4% a round trip. This already happened: from Feb to Jun 2024 the strategy made +0.7% against +17.8% for the universe, with 14 filter flips and 36% winning trades. Only 5% of the 800 settings on page 3 made over 10% in those months, so tuning does not fix it. **A less one-way market.** Apr 2023 to Jan 2024 gave 75% of the log return, when the universe alone scored Sharpe 5.65. Outside that stretch the strategy's Sharpe is 1.42 and its yearly return 16.8%. **Loss of leadership.** With no multi-baggers to catch, the top-10 share above suggests much of the return goes. **Corporate actions.** An event like ABFRL or STAR on a held name costs about 3% of equity at 1/15 weight, and nothing can see it coming. **Capacity.** The Rs 10 cr floor keeps small names eligible, and 0.2% a side may understate the cost of trading them in size.

Survivorship bias in the Task 2 universe

The list is the set of stocks liquid in Oct 2026, so membership looks into the future: all 555 names are still trading on the last date, and failures, mergers and flopped IPOs are missing. It shows up three ways. An equal-weight portfolio of everything made +148% before costs against +44.6% for Nifty, with only 11.4% of names falling. 15 random stocks score Sharpe 1.66 (std 0.36), so the firm's reference of 1.93 is under one standard deviation above chance. And the less liquid a stock was in 2022, the more it later rose, because a stock with almost no trading then is only here if something large happened to it.

Mean return to Jun 2024, by liquidity on 28 Feb 2022



Effect on the strategy. Raising the liquidity floor from Rs 10 cr to Rs 25, 50 and 100 cr takes Sharpe from 2.97 to 2.66, 1.92 and 1.64, and CAGR from 46.6% to 40.5%, 27.6% and 23.7%. That gap of about 19 points of CAGR and 1.0 of Sharpe at Rs 50 cr is my rough estimate of the flattery. It is a guide, not a bound: it includes any genuine small-cap effect, and it ignores the bias remaining above Rs 50 cr. Injecting up to 60 failing clones of real stocks barely moved the result (Sharpe 2.99 to 3.05), so the direct cost of holding failures is small, because losers exit quickly. The larger effect is inflated winners.

Measuring it properly. Rebuild a point-in-time universe from NSE bhavcopies, including later-delisted names, apply the same liquidity rule each day, and re-run. Failing that, compare the equal-weight universe with an index whose past membership is fixed, such as Nifty 500 Equal Weight. Details are in *survivorship_note.md*.

Would more tuning help?

No. I tested 800 settings per task with cross-validation (CSCV, Bailey et al.): pick the best on half the data, score it on the other half, and repeat for every split.

Sharpe	Task 1, all	Task 1, no 14:45	Task 2
Best settings, on the half used to pick them	3.78	3.46	3.72
Same settings, on the unseen half	2.97	2.35	2.84
Submitted settings	2.95	2.95	3.01
How often the best settings beat submitted	54%	5%	47%

I chose the submitted settings after seeing this data, so their row is flattered. The fair reading is the fall of about 1 Sharpe from picking winners, and I expect mine to fall on hidden data too.

Task 1 trade-off. Deciding at 14:45 gives Sharpe 3.57, worst loss -3.7% and 25.1% a year, a significant gain (95% CI +0.22 to +1.03). But a 2-bar delay then scores -3.37, because the fill misses the overnight gap. I kept 14:30 for that reason. **Task 2 alternatives.** None is significantly better (means of 4 offsets, against 2.97): a skip-month signal 3.27, 12 positions with sell rank 20 3.09, and breadth 0.70 3.33, which falls to 0.78 in the tough case.

Checks

No look-ahead. The harness check passes. On Task 1, 528 further truncation cuts gave no mismatches, and corrupting all prices after a cut changed nothing before it. Task 2 only ever sees *hist*.

Reproducible. Deterministic, with no dates in the code. Results are identical to 10 decimals on pandas 3.0.2 with numpy 2.4.4 and on pandas 2.2.3 with numpy 1.26.4. Task 1 runs in under 1 second and Task 2 in about 5 seconds, including on a test window as long as the hidden one.

Robust to odd data. Both handle short sessions and cold starts: with too little history they stay flat. Task 2 also handles missing prices on a held stock and injected price shocks.

What I looked at beyond the scored windows. The brief reserves 2021 for indicator warm-up. The scored runs trade nothing before the window, but in research I also ran Task 2 over Oct 2021 to Feb 2022 as a weak-market check, and that and the tough case informed its design, so neither is independent evidence. Task 1's later-period check uses the pack's daily Nifty file for Feb 2022 to Jun 2024, which overlaps the hidden window, so that stretch is not unseen by me.

Selection effects. 381 Task 1 and 1,396 Task 2 configs are logged. Designs were compared by averages across many configs, not by the best row. The same discipline is why I rejected the highest in-sample Task 2 threshold and why Task 2 tables average over rebalance offsets: moving the rebalance day alone shifts Sharpe by about 0.3.

Notes on the brief

Parameter limit. The email allows more than 6 parameters, but *check_params* rejects them. So PARAMS has 6 and the others are module constants, listed with their tests above. **Universe size.** The email says about 585 stocks, while the data and README have 555. **Undisclosed anomaly.** NMDC jumps +85.7% on 2022-10-27. The README lists only drops.

Other files. *failed_ideas.md* has what did not work, with numbers. Research code and all logged runs are at github.com/Prathama-1/Quant-Research, without the firm's data or harness.

With more time

Nifty futures bars. How much of the spot gap can a futures trade actually capture? This is the largest open question in Task 1, and the data would answer it directly.

A point-in-time stock universe. This is the largest open question in Task 2, as set out on the left.

A longer history. Task 1's filter has been tested on one crash and one recovery, and Task 2 on a window that is mostly a bull market. I would want both tested through a long bear market such as 2008 before trusting them with capital.